AWE NUCLEAR SECURITY TECHNOLOGIES

EPOCH Workshop 2026: Physics packages

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Current EPOCH physics packages

- Bremsstrahlung (e^-)
- Bremsstrahlung (e^+)
- Bethe-Heitler pair production (e^-e^+)
- Bethe-Heitler pair production ($\mu^-\mu^+$)*
- Background elastic-scatter collisions
- Nuclear-trident pair production*, **

- Dielectronic recombination
- Radiative recombination
- Binary elastic-scatter collisions (1 species)
- Binary elastic-scatter collisions (2 species)
- Electron-impact collisional ionisation
- 3-body recombination

- Non-linear Compton scatter
- Breit-Wheeler pair production
- QED-trident pair production**
- Field ionisation

Key:

- Particle interacts with species background
- Particle interacts with field background
- Particle interacts with local particles

* Not available in main release, only on dedicated feature branches
** Note: “trident” processes are unrelated to any AWE program

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Sampling theory: fixed rate process

dt: Small time interval

Δt: Long time interval ($\Delta t = ndt$)

λ: Event rate

k: Number of events over Δt

- Probability of event over dt: $P(dt)$
  - $P(dt)$=λdt
- Probability of no event over Δt: $\bar{P}(\Delta t)$
  - $\bar{P}(\Delta t) = \left(1 - P(dt)\right)^n = \left(1 - \frac{\lambda \Delta t}{n}\right)^n$
- As $dt \to 0$:
  - $\lim_{n \to \infty} \left(\bar{P}(\Delta t)\right) = e^{-\lambda \Delta t}$
- Probability of at least 1 event over Δt
  - $P_{k>0}(\Delta t) = 1 - e^{-\lambda \Delta t}$

Key equation:

$$P(\Delta t) = 1 - e^{-\lambda \Delta t}$$

(assuming $P_{k>1}(\Delta t)$ is negligible)

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Sampling theory: variable rate process

- Replace $\lambda \Delta t$ with $\tau = \int_0^{\Delta t} \lambda(t) dt$
- $\bar{P}(\tau) = e^{-\tau}$
  - This is a PDF, normalised from $\tau = 0 \rightarrow \infty$
- $\bar{F}(\tau) = \int_0^{\tau} e^{-\tau'} d\tau' = -e^{-\tau}$
  - CDF of survival to optical depth τ
- To sample an optical-depth for event-trigger τ_e, replace $\bar{F}(\tau)$ with x_r (random number from 0 to 1) and find τ:
  - $x_r = -e^{-\tau_e}$
  - $\tau_e = \ln(-x_r)$

Key equations:

$$\tau_e = \ln(-x_r)$$

Optical depth method:

1. Sample τ_e for each particle
2. Each step, subtract τ from τ_e
3. When $\tau_e < 0$, trigger event
4. Sample new τ_e if particle survives

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Compilation flags

- Several EPOCH features are locked behind compilation flags
  - These slow the code, only use if needed

- To activate:
  - Uncomment the DEFINES line in the Makefile
  - Run terminal command: make clean
  - Recompile the code: make COMPILER=gfortran

- Extra features:
  - Different macro-particle shapes
  - Physics packages
  - Extended IO (e.g. particle ID, probe-time)

```
# Use second order particle weighting (default is third order).
#DEFINES += $(D)PARTICLE_SHAPE_TOPHAT

# Use fifth order particle weighting (default is third order).
#DEFINES += $(D)PARTICLE_SHAPE_BSPLINE3

# Include a unique global particle ID. The first flag defines the ID using
# an 8-byte integer, the second uses 4-bytes.
#DEFINES += $(D)PARTICLE_ID
#DEFINES += $(D)PARTICLE_ID4

# Include QED routines
#DEFINES += $(D)PHOTONS

# Use the Trident process for pair production
#DEFINES += $(D)TRIDENT_PHOTONS

# Include bremsstrahlung routines
#DEFINES += $(D)BREMSSTRAHLUNG
```


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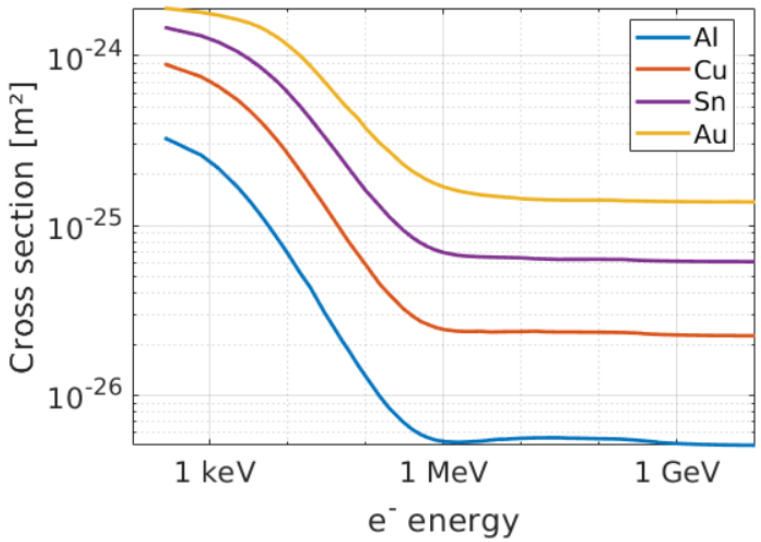
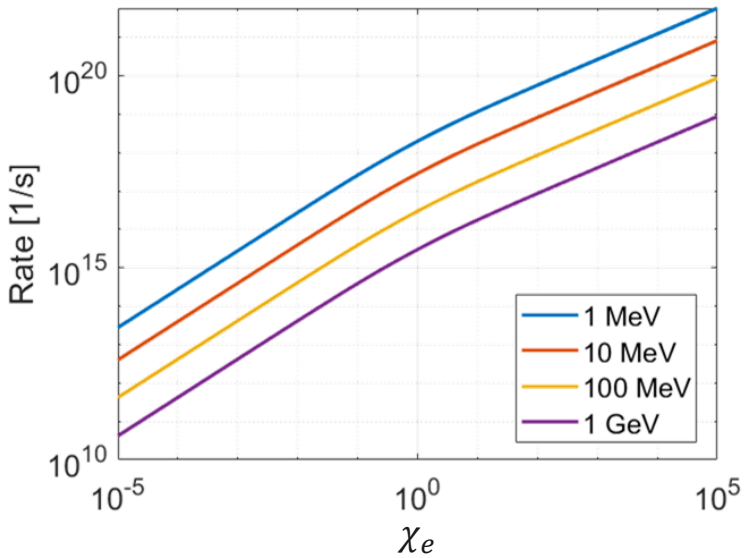
Radiation processes

- Two types of radiation:
  - Bremsstrahlung radiation
  - Non-linear Compton scatter (synchrotron radiation)

- Bremsstrahlung:
  - Trigger rate: $\lambda = n_i v_e \sigma$
  - σ: Cross-section (Seltzer-Berger tables)
  - n_i: Background ion density
  - v_e: Incident electron speed

- NCS (Ridgers 2014):
  - Trigger rate: $\lambda = \frac{\sqrt{3}\alpha_f c}{\lambda_c}\frac{\chi_e}{\gamma}h(\chi_e)$
  - χ_e scales with field-strength and electron energy





Morris *et al* (2021). *Physics of Plasmas*, *28*(10).

Ridgers *et al* (2014). *Journal of computational physics*, *260*, 273-285.

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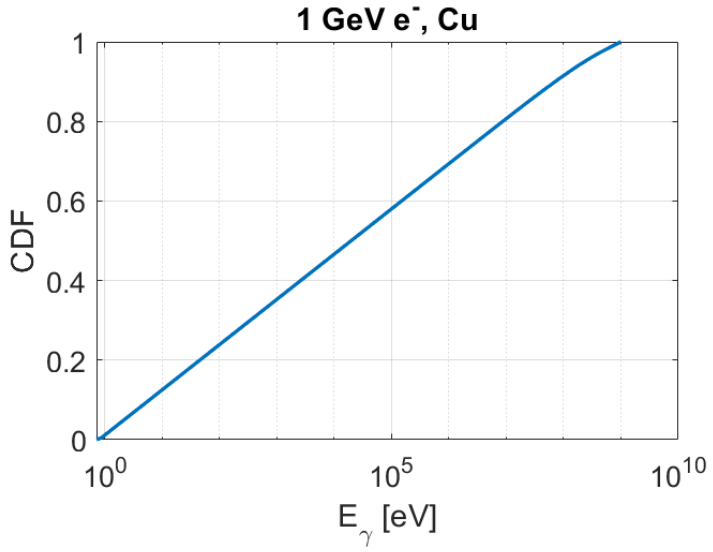
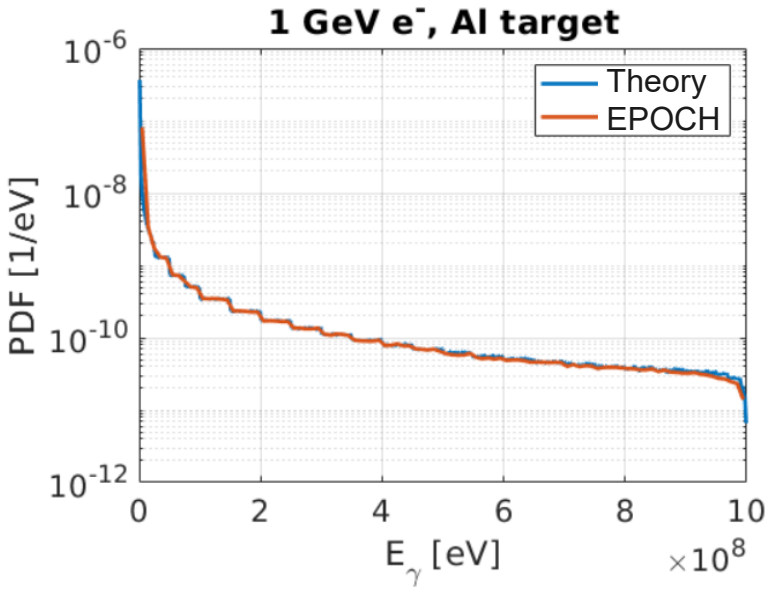
Macro-photon emission

- In photon emission:
  - Incident e^- has momentum reduced
  - Macro-photon created at e^- position
- Macro-photon energy:
  - Normalise $\frac{d\sigma}{dE_\gamma}$ to get photon emission energy distribution, PDF
  - Integrate PDF to get cumulative distribution function CDF
  - Replace CDF with a random number, return corresponding E_γ
- Photon angular distribution obtained from model fits
  - Usually parallel to incident e^- at high energy





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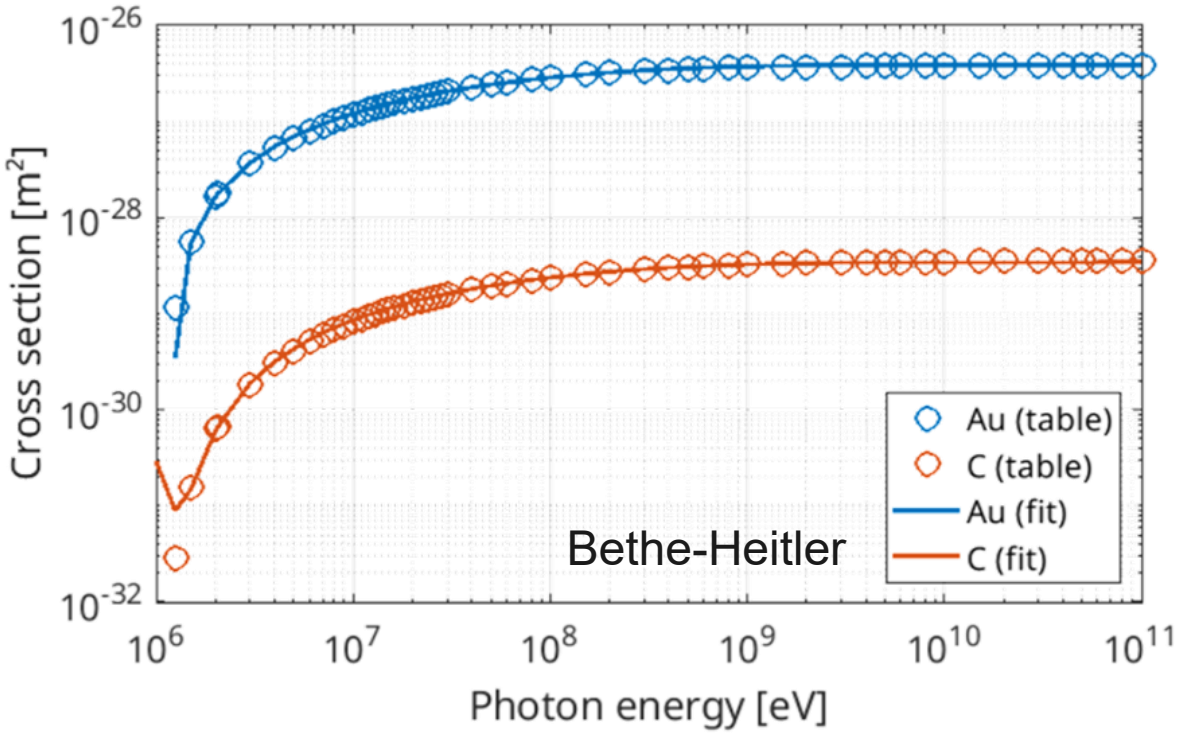
Pair production processes

- Main branch pair-production in EPOCH:
  - Breit-Wheeler (photon decay in laser fields)
  - QED-trident (e^- radiating an e^-e^+ pair in laser field)
  - Bethe-Heitler (photon decay in nucleus Coulomb field)

- Incident particle:
  - Destroyed (photon)
  - Recoiled (electron)

- Pair energies sampled using CDF method
  - CDF values read from tables
  - Create macro-electron and macro-positron



Martinez, *et al* (2019). *Physics of Plasmas*, *26*(10).
Ridgers *et al* (2014). *Journal of computational physics*, *260*, 273-285.

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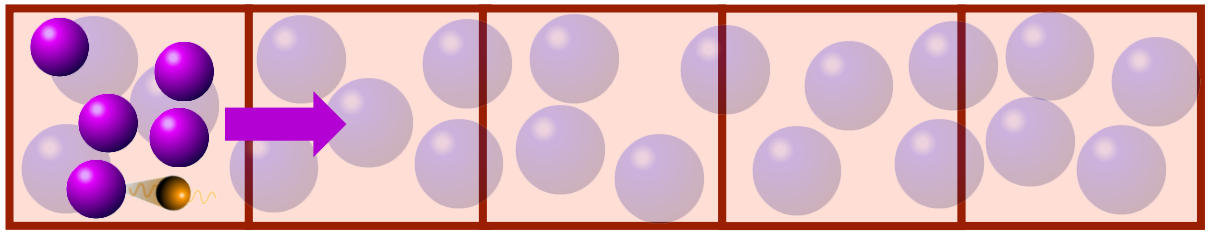
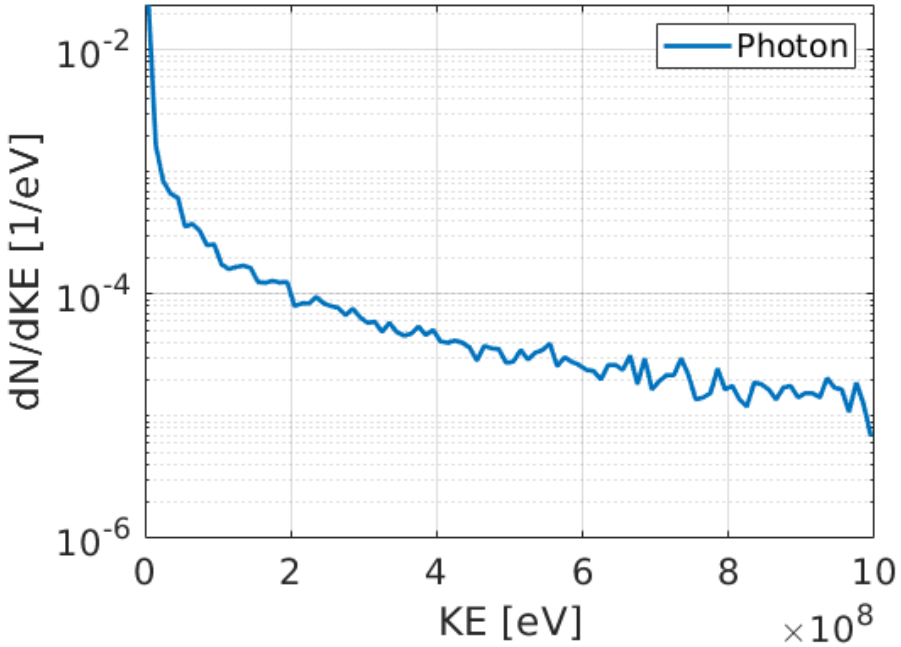
Example 4.1: Bremsstrahlung radiation

```
begin:bremsstrahlung
  use_bremsstrahlung = T
  start_time = 0
  produce_photons = T
  photon_energy_min = 1 * kev
  photon_weight = 1.0
  photon_dynamics = F
  use_bremsstrahlung_recoil = T
  use_plasma_screening = F
end:bremsstrahlung
```

- 1 GeV e^- passing through 5.5 mm of Al, over 20 ps
- Bremsstrahlung block sets physics properties:
  - Always recoil electrons when they radiate
  - If emitted photon energy > 1 keV, create a macro-photon
  - Option to increase emission number, reducing macro-weight (not used)
  - Do not move photons, let them sit at creation point
  - Ignore theoretical σ increase for heated targets (plasma-screening)





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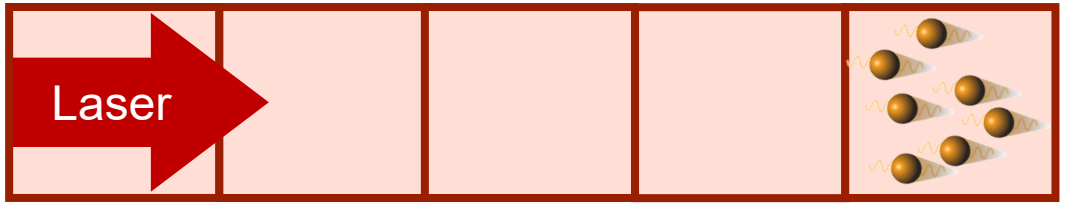
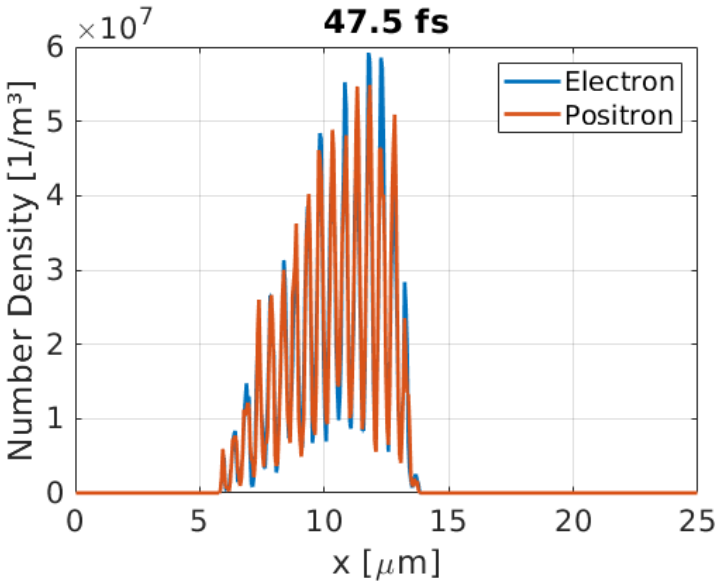
Example 4.2: Breit-Wheeler pair production



- 100 MeV photon bunch incident on a $2 \times 10^{23}\ \mathrm{Wcm}^{-2}$ laser

- qed block sets process properties:
  - Activate pair production
  - Allow photons to move
  - Also provides instructions for e^- NCS radiation, similar to brem.




```
begin:qed
  use_qed = T
  produce_photons = T
  photon_energy_min = 50 * kev
  produce_pairs = T
  photon_dynamics = T
end:qed

begin:boundaries
  bc_x_min = simple_laser
  bc_x_max = open
end:boundaries

begin:laser
  boundary = x_min
  intensity_w_cm2 = 2.0e23
  lambda = 1.0e-6
end:laser

begin:species
  name = Photon_Beam
  frac = 1
  rho = if (x gt 20.0e-6, 1.0e10, 0)

  drift_x = -100.0e6 * ev / c

  identify:photon
end:species
```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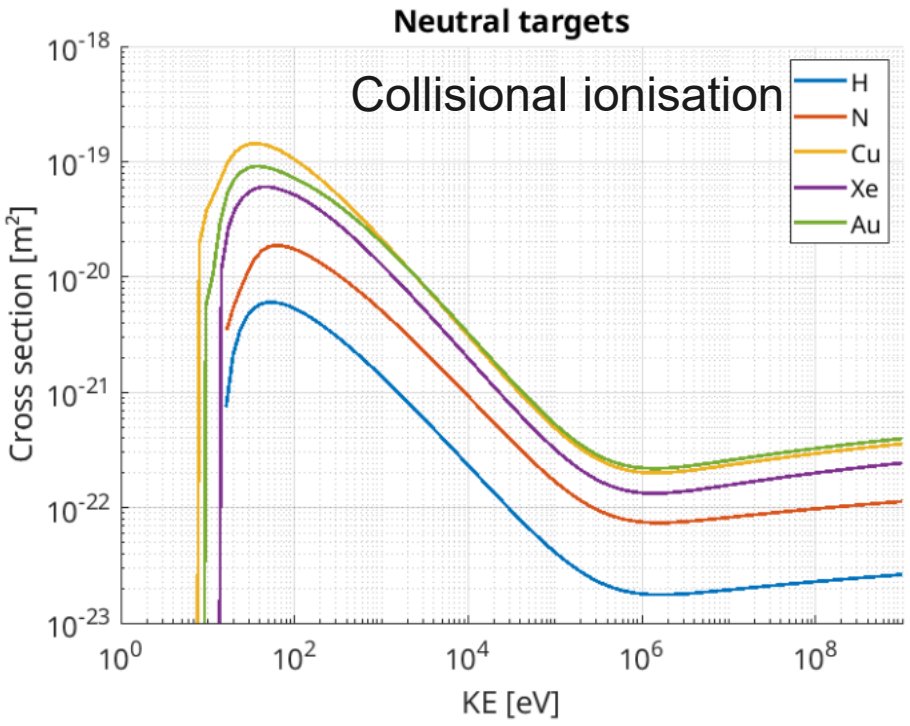
Ionisation/recombination processes

- Species identified by charge-state and atomic number
  - Ionisation/recombination rates calculated
  - Macro-ions move between different charge species
- Field ionisation:
  - Ion releases e^- in strong external field
- Collisional ionisation:
  - Incident e^- collides with bound e^-
  - Energy transfer in collision ejects bound e^-
- Recombination:
  - Incident e^- captured by ion



Morris *et al* (2022) *Physics of Plasmas*, *29*(12).

Lawrence-Douglas (2013). *Ionisation effects for laser-plasma interactions by particle-in-cell code* (PhD thesis, University of Warwick).

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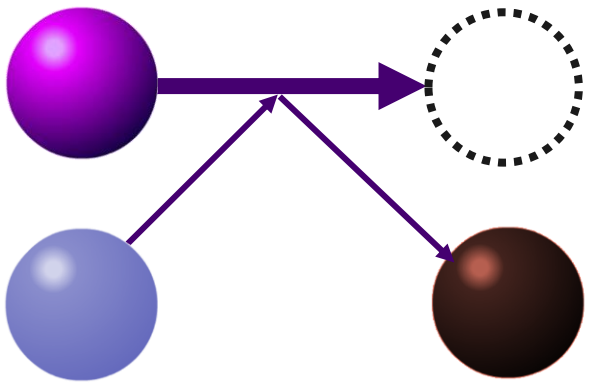
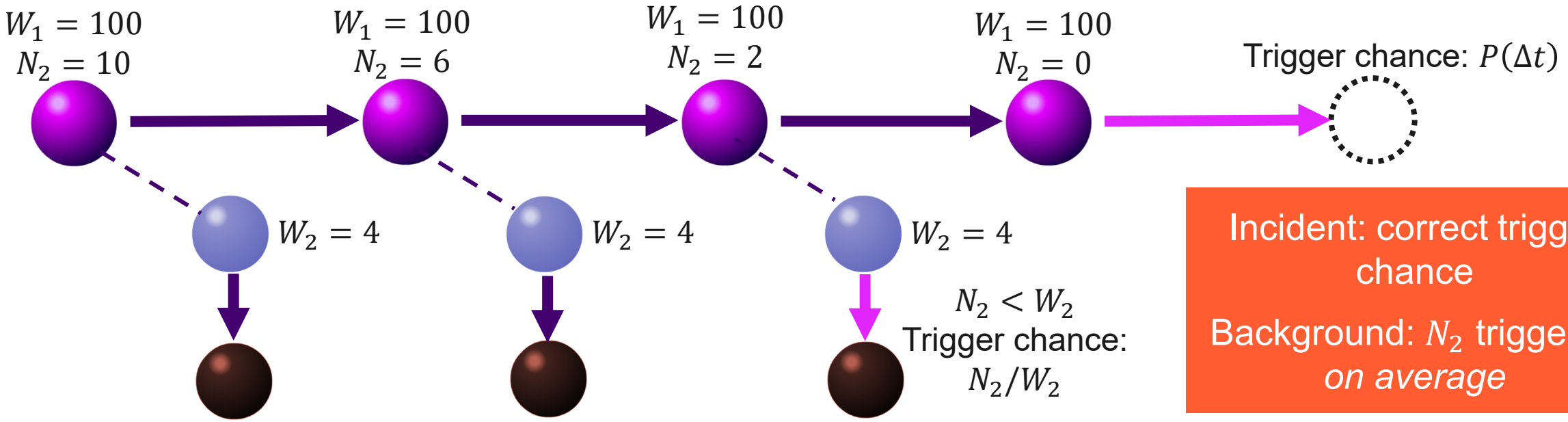
Collisional ionisation sampling

- Incident macro-particle represents W_1 real particles
- Each incident has interaction chance $P(\Delta t)$
  - Number of affected background particles: $N_2 = W_1 P(\Delta t)$
  - Trigger background macro-particles until N_2 are affected
- Trigger incident with chance $P(\Delta t)$ - e.g. 10%:





Incident: correct trigger chance

Background: N_2 triggered *on average*

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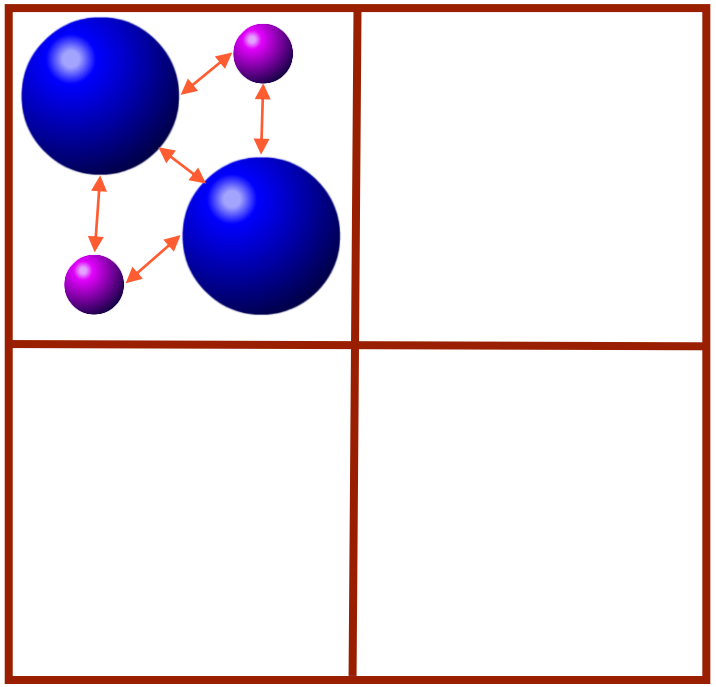
Collisions

- Fields can only be resolved between cells
  - What about local interactions?
- Separate package models collisions within the same cell
- Binary-collision algorithm:
  - Particles paired together
  - Scattering estimated and applied



Nanbu (1997). *Physical Review E*, *55*(4), 4642.
Pérez *et al* (2012). *Physics of Plasmas*, *19*(8).

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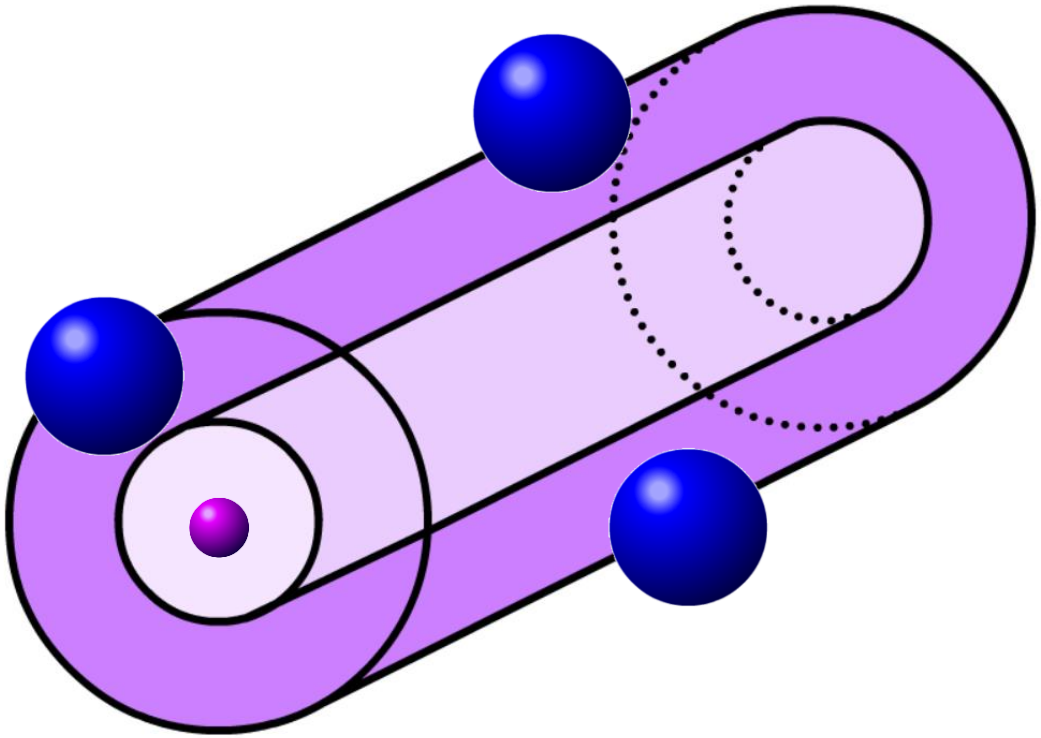
Nanbu numerical experiment

- Numerical experiment to determine scatter
- Randomly assign impact parameters inside cylinder
  - $b_{\max}$: Debye length λ_D
  - $b_{\min}$: Large angle scatter limit b_0
- Calculate cumulative scatter angles using single particle scatter:

$$\tan\frac{\theta_k}{2} = \frac{|q_\alpha q_\beta|}{4\pi\epsilon_0 \mu v_r^2 b}$$



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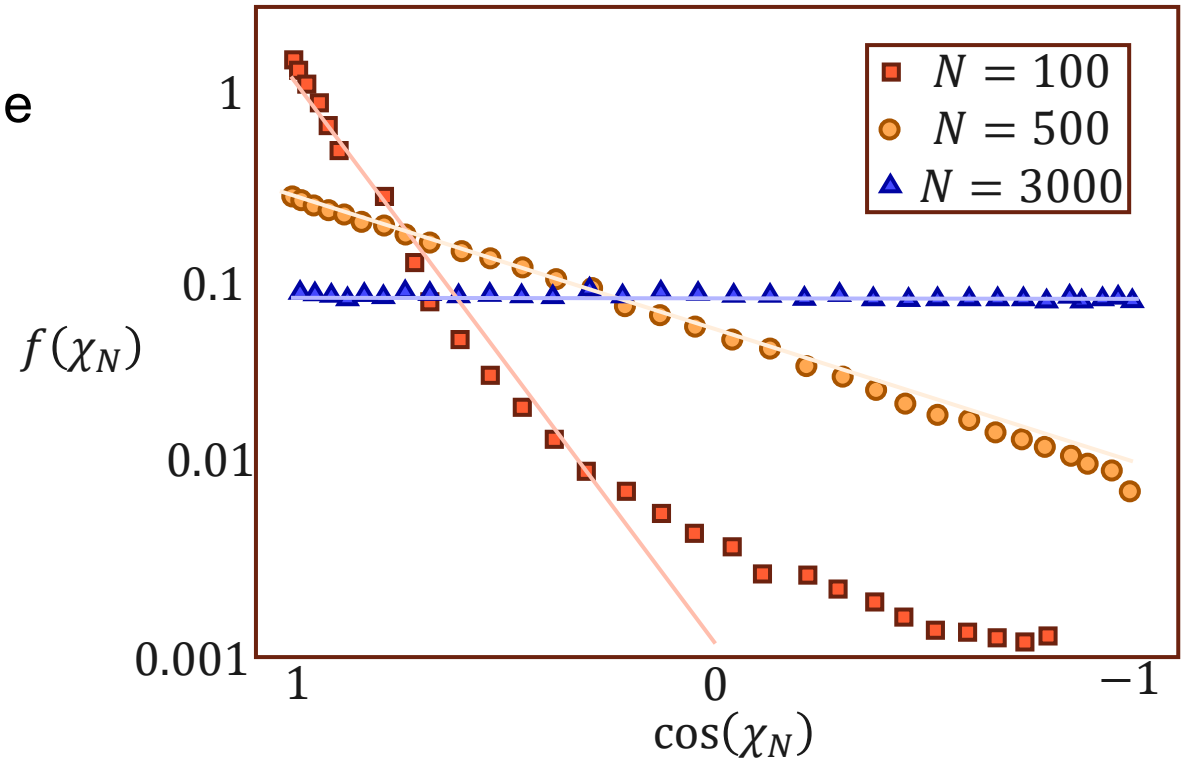
Nanbu-Pérez scatter

- $\cos\chi_N$: Cosine of cumulative scatter angle after N scatter events
- $\ln[f(\chi_N)]$:Probability distribution of scatter angle after N scatter events
  - Low N, peaked near small angle
  - High N, flat distribution: isotropic
- Apply linear fit to distributions
- In practice:
  - Weighted step: $s = n_\beta g \pi b_0^2 (\ln\Lambda)\Delta t$
  - Sampling term: $\coth A - A^{-1} = e^{-s}$
  - Angle: $\cos\chi = \frac{1}{A}\ln(e^{-A} + 2x_r \sinh A)$



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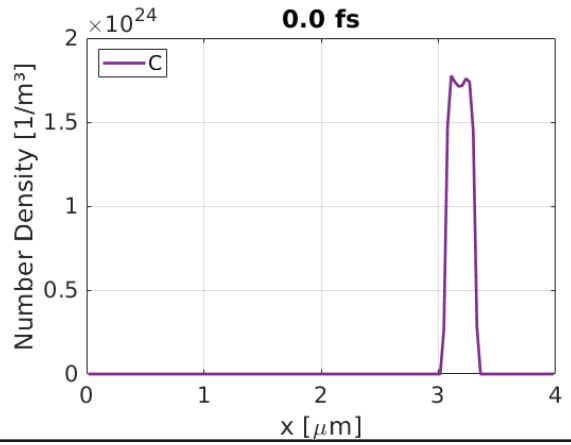
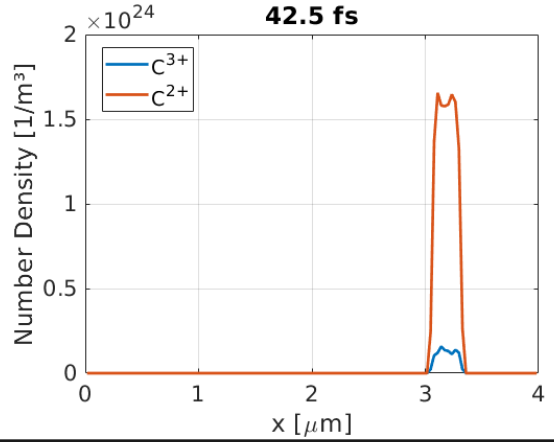
Example 4.3: Field ionisation

- Shoot a $3 \times 10^{15}\ \mathrm{Wcm}^2$ laser at a block of neutral C
  - Without field ionisation, the laser would pass through the target
  - Switch on field ionisation in control block
- Set up Carbon species for ionisation
  - ionise: Mark Carbon as an ionisation base-species
  - ionise_limit: Number of ionised species to create (up to C^{3+} here)
  - unique_electron_species: Make a separate e^- species for each ionised charge-state






```
begin:control
  nx = 128
  npart = 20 * nx
  t_end = 42.4e-15
  x_min = 0
  x_max = 4.0e-6
  field_ionisation = T
end:control

begin:boundaries
  bc_x_min = simple_laser
  bc_x_max = open
end:boundaries

begin:laser
  boundary = x_min
  intensity = 3.0e15 * 1.0e4
  lambda = 800.0e-9
  t_profile = 1
  t_end = 10.7e-15
end:laser

begin:species
  name = Carbon
  charge = 0
  atomic_no = 6
  mass = 1836.2 * 16.0
  frac = 1
  rho = if ((x lt 3.30e-6) and (x gt 3.05e-6), 1.74e24, 0)
  ionise = T
  ionise_limit = 3
  unique_electron_species = T
end:species

begin:output
  dt_snapshot = t_end/100
  number_density = always + species
end:output
```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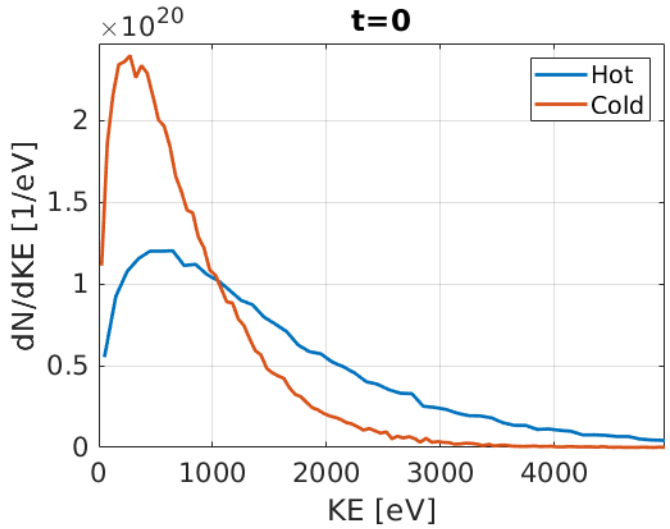
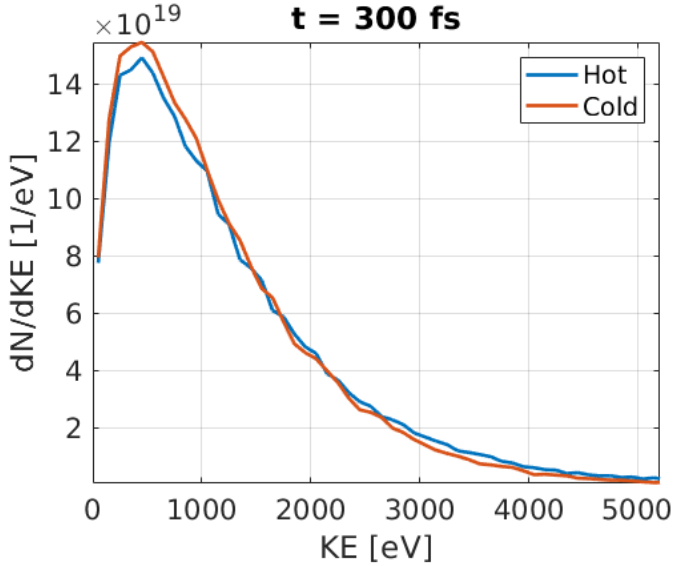
Example 4.4: Collisional equilibration

- Start with two overlapping e^- species
  - Initial temperatures: 1 keV, 500 eV
  - Switch on collisions, watch thermal equilibration

- Collisions:
  - collide: Identify which species pairs can collide
  - coulomb_log: Set to calculate $\ln(\Lambda)$ in each cell, each step. Very slow – much better to give an approximate value (e.g. 5)
  - coll_n_step: Run collisions once every n steps, for timestep $n\Delta t$. Speed-up technique for boosting collisions number

```
begin:collisions
   use_collisions = T
   use_nanbu = T
   coulomb_log = auto
   collide = all
   coll_n_step = 10
end:collisions
```





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